

PROJ. NUM.	PROJECT NAME	EMPLOYEE NUMBER	EMPLOYEE NAME	JOB CLASS	CHG/HOUR	HOURS BILLED	TOTAL CHARGE
15	Evergreen	103	June E. Arfoughi	Elec. Engineer	\$ 85.50	23.8	\$ 2,011.70
		101	John C. News	Database Designer	\$105.00	19.4	\$ 2,037.00
		105	Alice K. Johnson*	Database Designer	\$105.00	35.7	\$ 3,748.50
		106	William Smithfield	Programmer	\$ 35.75	12.6	\$ 450.45
		107	David H. Senior	Systems Analyst	\$ 96.75	23.8	\$ 2,302.65
				Subtotal			\$10,549.70
18	Amber Wave	114	Annelise Jones	Applications Designer	\$ 48.10	25.6	\$ 1,183.26
		118	James J. Frommer	General Support	\$ 18.36	45.3	\$ 831.71
		104	Anne K. Ramoras*	Systems Analyst	\$ 96.75	32.4	\$ 3,135.70
		112	Darlene M. Smithson	DSS Analyst	\$ 45.95	45.0	\$ 2,021.80
				Subtotal			\$7,172.47
22	Rolling Tide	105	Alice K. Johnson	Database Designer	\$105.00	65.7	\$ 6,793.50
		104	Anne E. Ramoras	Systems Analyst	\$ 96.75	48.4	\$ 4,682.70
		113	Delbert K. Jacobbrood*	Applications Designer	\$ 48.10	23.6	\$ 1,135.16
		111	Geoff B. Wahash	Clerical Support	\$ 26.87	22.0	\$ 591.14
		106	William Smithfield	Programmer	\$ 35.75	12.8	\$ 457.60
				Subtotal			\$13,660.10
25	Starflight	107	Maria D. Alonzo	Programmer	\$ 35.75	25.6	\$ 915.20
		115	Travis B. Bawangi	Systems Analyst	\$ 96.75	45.8	\$ 4,431.15
		101	John C. News*	Database Designer	\$105.00	56.3	\$ 5,911.50
		114	Annelise Jones	Applications Designer	\$ 48.10	33.1	\$ 1,592.11
		108	Ralph B. Washington	Systems Analyst	\$ 96.75	23.6	\$ 2,283.30
		118	James J. Frommer	General Support	\$ 18.36	30.5	\$ 559.98
		112	Darlene M. Smithson	DSS Analyst	\$ 45.95	41.4	\$ 1,902.33
				Subtotal			\$17,559.82
				Total			\$48,942.09

Note: * indicates project leader.

Important Note :

 Highlighted Text → PK

 Underlined Text → FK

0 Normal Form :

Project (**Project_Num** , Project_Name , Employee_Number ,
Employee_Name , Job_Class , CHG/HOUR , Hours_Billed , Total_Charge)

1. 1NF — First Normal Form

- A relation must contain **no Multivalued Attributes**.
- No **Repeating Groups**.
- No **Composite Attributes**.
- To achieve 1NF:
 - **Repeating Group:** Move it to a new table + add the original PK as a FK.
 - **Multi-valued Attribute:** Move it to a new table + add the original PK as a FK.
 - **Composite Attribute:** Split its subparts into separate columns.

Project (**Project_Num** , Project_Name)

Project_Employee (**Project_Num** , **Employee_Number** , Employee_Name ,
Job_Class , CHG/HOUR , Hours_Billed)

2. 2NF — Second Normal Form

- Must be in **1NF**.
- No **Partial Dependency**.
- Every Non-Key Attribute must depend on the **whole PK**.
- To achieve 2NF:
 - Move the partially dependent Non-Key Attributes to a new table.
 - Add the **part of the PK they depend on** as the new table's PK/FK.

Project (**Project_Num** , Project_Name)

Project_Employee (**Project_Num , Employee_Number** , Hours_Billed)

Employee (**Employee_Number** , Employee_Name , Job_Class , CHG/HOUR)

3. 3NF — Third Normal Form

Must be in 2NF.

- No **Transitive Dependency**.
- A Non-Key Attribute must not depend on another **Non-Key Attribute**.
- To achieve 3NF:
 - Move the dependent Non-Key Attributes to a new table.
 - Keep the Non-Key Attribute they depend on in the new table.
 - Keep that determining Non-Key Attribute in the original table as well.

Project (**Project_Num** , Project_Name)

Project_Employee (**Project_Num** , **Employee_Number** , Hours_Billed)

Employee (**Employee Number** , Employee_Name , Job Class , CHG/HOUR)

Job_Class (**Job_Class** , CHG/HOUR)